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Innovative guidelines for mix design and construction practices for concrete manufactured using rice husk ash and waste marble slurry: insights from experimental study

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Abstract

This study investigates the use of rice husk ash (RHA) and waste marble powder (WMS) as sustainable pozzolanic additives in concrete, promoting circular economy principles by reusing the wastes. Cement was partially replaced with RHA (6%–24%) and fine aggregate (FAA) with WMS (12%–36%), alongside 1.5% hooked steel fibres. The optimized mix, RH18MD36SF1.5 (18% RHA, 36% WMS, 1.5% steel fibres), showed significant improvements: compressive strength increased by 47.6%, split tensile by 45.8%, and flexural by 49.6%. Life cycle assessment (LCA) using the ReCiPe methodology revealed reductions of 36.5% in particulate emissions, 36.6% in greenhouse gases, and 41.7% in fossil fuel burning. These results align with UN SDGs 9 and 12, promoting climate-resilient, sustainable infrastructure. The study demonstrates the feasibility of converting waste into construction resources, offering guidelines for optimizing concrete mix designs and construction practices to enhance durability and sustainability in structural applications.

1. Introduction

Concrete production is assumed to reach 18 billion tons by 2050 which poses a severe threat to environment considering the energy consumed and pollution generated by concrete owing to its non-biodegradability [1]. The cement in concrete binds aggregates, but its production releases enormous amounts of CO₂ leading to global warming [2]. Some of the pozzolanic materials represent such sustainable alternatives to conventional cement which, though not having inherent binding properties, develop cementitious qualities in the presence of water [3].

RHA is created from agricultural waste emanating from rice cultivation estimated at 499.31 million metric tons [4], and 0.28 kg of Rice husk is produced for every kg of rice [5]. Rice Husk Ash (RHA) has been considered an effective cement replacement in concrete with additional advantages economically and environmentally friendly advantages [4]. RHA is known for its pozzolanic properties, providing strength improvement and environmental benefits [5]. In contradiction, some studies reported reduced workability with minimal strength increase due to use of RHA in concrete [6]. This could be attributed to its high silica content, which can lead to insufficient pozzolanic reaction if not properly processed, and its fine particle size, which increases water demand in the mix.

In fact, optimum cement replacement with RHA has resulted in improved compressive strength (f_{cts}) according to research; the enhancement is quite significant at 10–15% cement replacement levels [7].

There are also several researchers reporting on the benefits of RHA addition in enhancing the concrete mechanical properties, especially for cement replacement levels of 10%, 15%, and 20%, where enhanced f_{cts} were developed due to quicker pozzolanic reactions [8]. The combination of RHA with some other materials like metakaolin has also been studied, which shows gain up to a certain replacement level [9]. Other authors have mentioned that small RHA additions improve pore filling and, in addition, results in enhanced production of C-S-H gel, which improves f_{cts} [8]. Several researchers have documented the strengthening effect of RHA in

concrete, and an increase up to 15% cement replacement is reported [7], while others show reduced workability or very minimal gains in strength when a higher percentage of cement is replaced by RHA [6].

The inclusion of RHA significantly reduced chloride penetration [10], hence improving the durability of concrete mixes. This is mainly because of the micro-filling ability of RHA, which reduces porosity [11, 12]. Also, addition of supplementary materials like metakaolin and polypropylene fibers along with RHA results in high compressive strength (f_{cs}) [9].

Apart from RHA, supplementary cementitious materials like WMS which is generated from processing of marble stones causes considerable environmental degradation and requires immediate attention to avert an environmental disaster [13]. As a solution, Fine aggregate (FAA) can be replaced with WMS in specified percentage which can lead to strengthening of the concrete because of its finer particles, which allows micro-filling effects leading to overall increase in the density of cementitious matrix [13]. The inclusion of WMS for replacement of FAA upto 50% significantly increases the concrete mechanical properties [14] and leads to subsequent environmental benefits [13]. However, substitution of FAA by WMS in percentage above 50% leads to significant loss of concrete strength and workability [15].

However, combination of RHA & WMS can have catastrophic effect on the environment. Improper handling of RHA can generate fine particulate dust, posing respiratory health risks to workers. Additionally, the disposal of untreated RHA in large quantities may lead to soil and water contamination due to its alkaline nature. Also, Waste marble powder (WMS) can have a negative environmental impact if not properly managed, leading to land degradation, air pollution due to fine dust, and potential contamination of water bodies [13]. As a potential solution, these products can be utilized to replace cement and FAA in concrete aligning with sustainable development goals [13]. The first main aim of the study is to understand the mechanism of reaction of RHA & WMS with other concrete components. Secondly, this study aims to determine the impact of replacement of cement with RHA and FAA with WMS on concrete properties. Thirdly, this study uses Recipe midpoint and endpoint analysis to quantify the environmental impact of using RHA & WMS on concrete which is missing in the previous literature survey.

The objectives of this study are to enhance the mechanical properties and sustainability of concrete by incorporating rice husk ash (RHA) and waste marble powder (WMS) as pozzolanic additives, while reducing environmental impact through circular economy practices.

The new concept in this research work is the synergistic integration of rice husk ash (RHA) and waste marble powder (WMS) as sustainable pozzolanic additives, combined with steel fibers, to simultaneously enhance mechanical properties, reduce environmental impact, and align with circular economy principles. Additionally, the study incorporates a detailed life cycle environmental impact assessment (EIA) using the ReCiPe methodology, providing a comprehensive framework for optimizing eco-friendly and high-performance concrete mix designs.

2. Experimental program

In the present investigation, ordinary Portland cement (OPC) as per BIS: 8112–2013 standard has been adopted [16] and OPC properties based on laboratory test are shown in table 1. The detailed composition of Ordinary Portland cement and WMS has been elucidated in table 2. Crushed gravel (20 mm and 10 mm size) is adopted as coarse aggregate (CAA) while fine river sand adhering to BIS 383–1970 standard [17] has been adopted as the FAA as shown in table 3. The sieve analysis results of FAA clearly shows that FAA lies in Zone 2 in accordance with IS10262:2009 [18] specifications as elucidated in figure 1. The properties of FAA & CAA has been elucidated in table 3. XRD study clearly indicated that dolomite is the major crystalline phase in WMS though the peaks corresponding to silicon dioxide and calcium carbonate were also present. WMS has been obtained from marble processing plants from Rajasthan and has been used to replace the FAA at levels ranging from 12 to 36%. **WMS was obtained from the cutting the marble slabs with diamond-edged blades and subsequently settling them in sedimentation tanks for 24 h followed by drying in sunlight for 3 days.** The sieve analysis results of WMS is described in figure 1 and chemical composition of WMS used by different authors has been described in table 4. Likewise, chemical composition and XRD results for RHA has been shown in table 5 and figure 2 respectively.

3. Mix proportions adopted

Mix proportions have been designed as per BIS 10262: 2009 [18]. Cement was replaced partially with RHA at variant levels from 6% to 24%, whereas FAA was replaced by WMS from 0% to 36% at a step size of 12%. Hooked steel fibers (HEF) were added at 1.5% by weight of cement. The concrete mixtures used in the experimental study has been categorized into four groups: one without RHA substitution, labelled 'RHA 0%', the second with cement replaced by 6% RHA, labelled 'RHA 6%', the third group with cement replaced by 12%

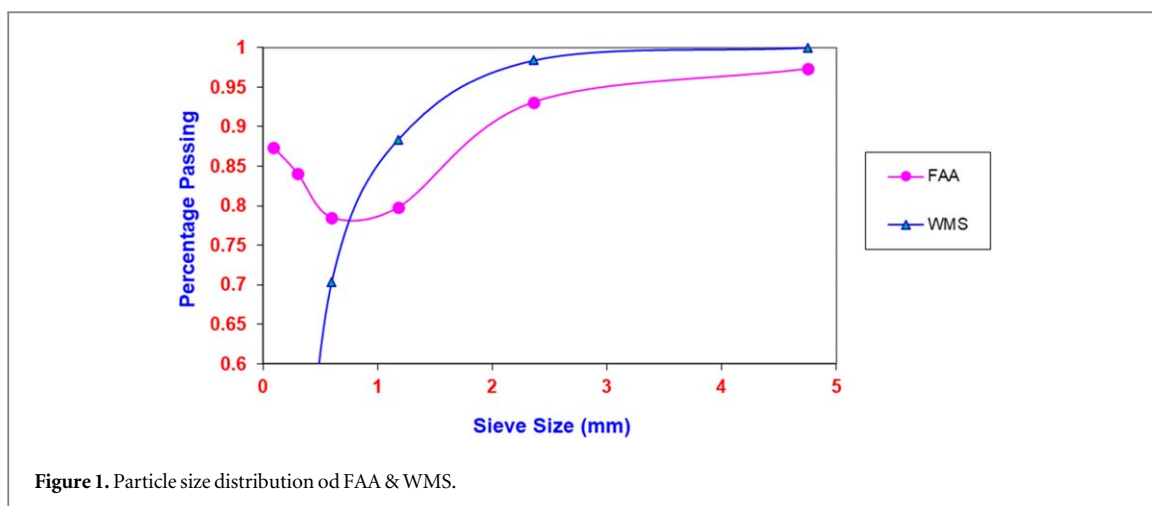


Figure 1. Particle size distribution of FAA & WMS.

Table 1. Experimental properties of OPC.

Tests	Experimental value	Standards as per IS 8112–2013
Normal consistency	29%	—
Initial Setting time (min)	135	Not less than 30 min
Final setting time (min)	280	Not greater than 600 min
Fineness (m^2/kg)	249.50	—
Specific gravity	3.14	—
f_{cts} in MPa		
3 days	33.42	22
7 Days	43.47	33
28 days	49.23	43

Table 2. Chemical composition of cement and WMS (IS 8112:2013).

Chemical composition	Cement	WMS
Calcium Oxide (CaO)	64.25	42.36
Silicon di Oxide (SiO ₂)	23.89	5.24
Aluminium Oxide (Al ₂ O ₃)	6.02	0.47
Iron Oxide (Fe ₂ O ₃)	2.49	0.72
Sulphur Trioxide (SO ₃)	1.22	0.13
Magnesium Oxide (MgO)	0.85	16.73
Potassium Oxide (K ₂ O)	0.48	0.081
Sodium Oxide (Na ₂ O)	0.8	0.054

Table 3. Physical properties of coarse aggregate (CAA) & fine aggregate (FAA).

Property	CAA		FAA		RHA
	20 mm	10 mm	Sand	WMS	
Specific Gravity	2.68	2.79	2.54	2.66	—
Water absorption (%)	0.87	0.98	1.64	1.89	—
Fineness modulus	6.94	6.54	2.56	2.48	<45 μm

RHA, labelled 'RHA 12%', In fourth and fifth group cement is replaced with 18% and 24% RHA as shown in table 6. Table 6 presents the mixture composition with steel fibers; the same composition without steel fibers was also used in the experimental study. To avoid redundancy, the composition is mentioned only once.

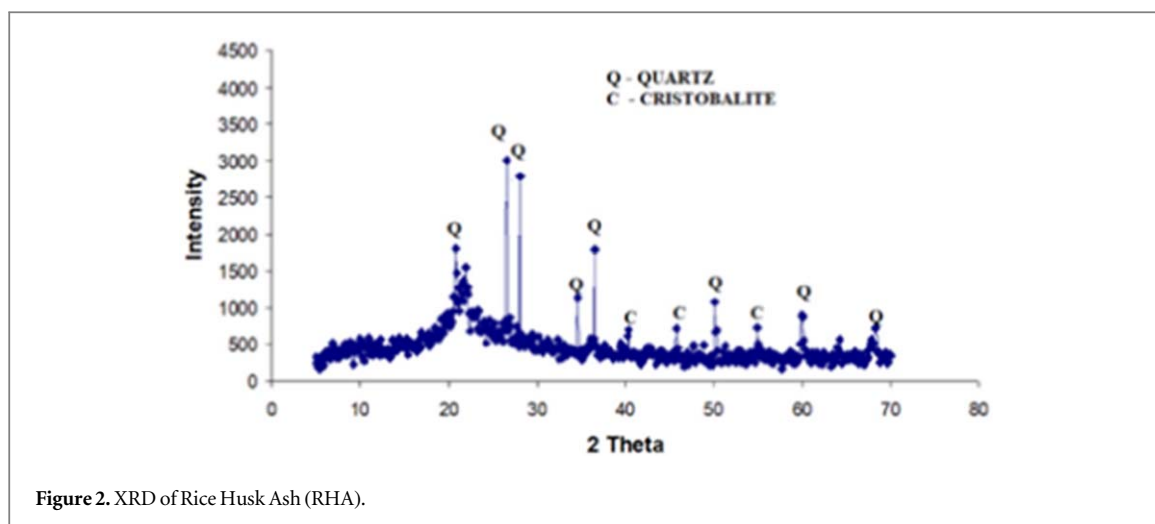


Table 4. Composition of WMS by different authors (Varadharajan *et al* 2020, Reference 13).

Reference Number	SO ₃ %	MgO%	CaO%	Fe ₂ O ₃ %	Al ₂ O ₃ %	SiO ₂ %	Other parameters
73	0.33	14.36	61.8	0.65	0.67	8.38	—
37	—	19.85	32.19	1.18	0.18	1.57	LOI –45.07
75	0.56	0.52	83.22	0.05	0.73	1.12	—
76	—	—	—	—	—	—	—
77	0.1	0.1	58.1	0.2	0.1	0.8	Na2O Nd, K2O—0.1, p205ND
78	—	8.74	1.58	2.33	4.45	64.86	—
79	—	2.77	42.14	1.94	2.69	14.08	—
36	—	1.03	54.86	0.04	0.08	0.15	—
80	—	16.25	40.45	9.7	0.42	28.35	—
81	—	4.17	54.55	0.32	—	0.14	—
82	—	8.74	1.58	11.99	4.45	64.86	—
83	3.28	1.54	63.61	3.53	4.83	18.95	—
84	—	2.56	4.83	6.54	18.75	62.48	—
85	0.02	18.94	32.23	1.09	1.09	4.99	LOI – 40.63, K2O—0.91NA2O 0.63
86	0.59	54.43	0.08	0.12	0.67	LOI 43.4	—
38	—	16.25	40.45	9.70	0.42	28.35	—
87	0.08	0.4	51.7	0.44	0.67	0.18	—

Table 5. Chemical composition of Rice Husk Ash (RHA).

Calcium Oxide (CaO)	3.4
Silicon di Oxide (SiO ₂)	92.3
Aluminium Oxide (Al ₂ O ₃)	1.9
Iron Oxide (Fe ₂ O ₃)	0.4
Sulphur Trioxide (SO ₃)	0.4
Magnesium Oxide (MgO)	0.94
Potassium Oxide (K ₂ O)	0.75
Sodium Oxide (Na ₂ O)	0.9

4. Compressive strength (f_{cts}) results

The pictures of experimental study are elucidated in figure 3. f_{cts} was determined as per procedure described in IS 4031 [19] by testing cubes of shape 150 mm × 150 mm × 150 mm, and the results are represented in figures 4, 5. Compressive strength was determined as per IS 4031 by preparing standard concrete cubes (150 mm × 150 mm × 150 mm) and curing them for the specified period. The cubes were tested using a compression testing machine by applying a load uniformly at a rate of 140 kg cm⁻² per minute until failure. The maximum load at failure was recorded to calculate compressive strength using the formula Strength = Load/ Area. Optimal concrete curing requires maintaining moisture and temperature (20 °C–25 °C) for proper

Table 6. Composition of concrete mixes for 1 cubic meter of concrete.

Mixture details	Cement (kg)	FAA (kg)	CAA (kg)	RHA (kg)	WMS (kg)	Steel Fibre (%)	W/C
RH0MD0SF1.5	480	620	1280	0	0	1.5	190
RH0MD12SF1.5	480	545.6	1280	0	74.4	1.5	190
RH0MD24SF1.5	480	471.2	1280	0	148.8	1.5	190
RH0MD36SF1.5	480	396.8	1280	0	223.2	1.5	190
RH6MD0SF1.5	451.2	620	1280	28.8	0	1.5	190
RH6MD12SF1.5	451.2	545.6	1280	28.8	74.4	1.5	190
RH6MD24SF1.5	451.2	471.2	1280	28.8	148.8	1.5	190
RH6MD36SF1.5	451.2	396.8	1280	28.8	223.2	1.5	190
RH12MD0SF1.5	422.4	620	1280	57.6	0	1.5	190
RH12MD12SF1.5	422.4	545.6	1280	57.6	74.4	1.5	190
RH12MD24SF1.5	422.4	471.2	1280	57.6	148.8	1.5	190
RH12MD36SF1.5	422.4	396.8	1280	57.6	223.2	1.5	190
RH18MD0SF1.5	393.6	620	1280	86.4	0	1.5	190
RH18MD12SF1.5	393.6	545.6	1280	86.4	74.4	1.5	190
RH18MD24SF1.5	393.6	471.2	1280	86.4	148.8	1.5	190
RH18MD36SF1.5	393.6	396.8	1280	86.4	223.2	1.5	190
RH24MD0SF1.5	364.8	620	1280	115.2	0	1.5	190
RH24MD12SF1.5	364.8	545.6	1280	115.2	74.4	1.5	190
RH24MD24SF1.5	364.8	471.2	1280	115.2	148.8	1.5	190
RH24MD36SF1.5	364.8	396.8	1280	115.2	223.2	1.5	190

hydration, typically for 7–28 days. High humidity (>90%) ensures strength development and durability. Common methods include water curing, membrane curing, and steam curing.

The mixing process was carried out using a concrete mixer to ensure uniform blending of cement, rice husk ash, aggregates, and water. Dry components were first mixed, followed by gradual addition of water for proper hydration. Stirring was performed using a vibration table, where the concrete was poured into moulds and compacted through vibration to eliminate air pockets and enhance density. This combination ensured a homogeneous mix, reduced porosity, and improved the structural integrity of the concrete products. The average mixing time was kept between 6–8 min. This is because, increased mixing time ensures uniform dispersion of fine RHA and WMS particles, preventing segregation due to their low densities. It also facilitates better integration with cement and aggregates, improving the homogeneity of the mix. This results in consistent workability and enhanced performance of the concrete.

From figures 4 and 5, it can be clearly seen that 28 day f_{cts} increased by **7.8%, 16.2%, and 22.8%** when cement was substituted by **6%, 12%, and 18%** RHA respectively. The strength increase is because of higher reactive silica in RHA, resulting in increased pore refinement and concrete matrix densification. Moreover, the release of entrapped air aggravates the hydration process resulting in strong interfacial transition Zone. However, strength enhancement did not increase comprehensively at 24% RHA replacement level and a drastic decrement of **15.3%** was observed in comparison to **18%** RHA substitution. This is because of the excess silica content which causes shortage of dicalcium silicate resulting in reduction in strength gain. The experimental study results are in close agreement with previous research works with Sai *et al* 2012 observing increase in f_{cts} due to cement replacement with RHA [20]. Likewise, Kannan and Ganesan 2013 reported enhanced f_{cts} till 15% RHA substitution level [21]. Thereafter, authors observed significant decrease in f_{cts} [14]. Similar findings were reported in previous research studies [22].

The experimental results further elucidated that substitution of FAA with WMS comprehensively increased f_{cts} , as depicted from figures 4 and 5. In fact, at 28 days, the f_{cts} increased by **4.7%, 8.6%, and 15.8%** for **12%, 24%, and 36%** substitutions of FAA with WMS, respectively in comparison to the reference mixture. This result is in agreement with previous research works as shown in table 7. This could be attributed to the fact that WMS have a fine particle size in range of 1.18 mm to 300 μ m which results in microfilling effect enhancing microstructure of concrete matrix [13]. The image reveals a compact and dense microstructure, indicating a significant reduction in voids and porosity, which directly contributes to improved strength [23]. Additionally, the presence of needle-like or plate-like structures suggests the formation of hydration products, such as calcium silicate hydrates (C-S-H), known for their role in enhancing the mechanical properties of cementitious materials. The uniform distribution of WMS particles within the matrix without noticeable agglomeration demonstrates effective mixing, which ensures a homogeneous structure and further strengthens the material. Improved bonding at the interfacial transition zones (ITZ) between WMS particles and the surrounding matrix is also evident, highlighting the role of WMS in reinforcing the material's integrity. Furthermore, the reduced presence of microcracks in the SEM image, compared to typical control samples, reflects enhanced structural



c) Broken Concrete Cube

Figure 3. Concrete mixing and testing.

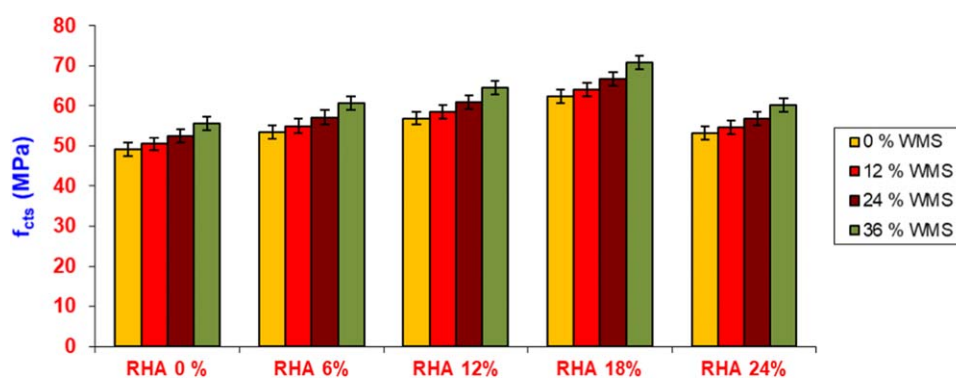


Figure 4. f_{cts} for concrete mixes without steel fiber.

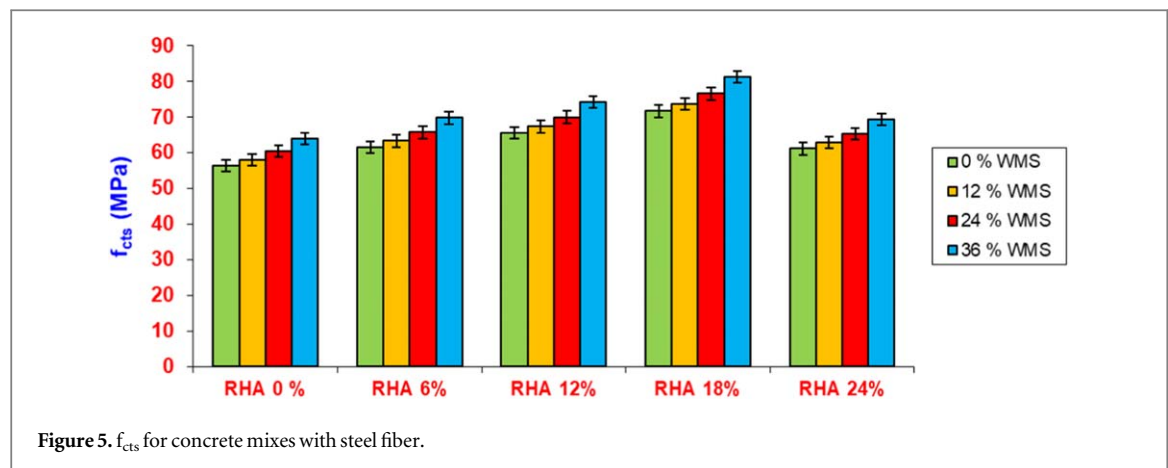


Table 7. (a) Experimental results from previous studies (Varadharajan *et al* 2020, Reference 13). (b) Comparison of use of RHA in concrete with previous research works (Varadharajan *et al* 2020, Reference 7).

Reference Number	Percentage of replacement (%)	Maximum percentage increase (%) in mechanical properties (red color indicates decrement of strength) f_{cts}	f_{spt}	f_{fpt}	Workability
(a)					
73	0, 10, 15, 10+10	15.88	27.78	—	Reduced
37	0, 20, 40, 60, 80, 100	91	86.6	—	No change
74	0, 25, 50, 75, 100	150	—	—	Reduced by 6%
15	10, 20, 30, 40, 50, 60	20	—	—	Decreases with 40% replacement
76	0, 20, 50, 100	—	—	—	Decreased
78	50% marble+50% quarry dust	7.7	25	—	Increase
79	5, 10, 15	15	10	—	Decreased
43	5, 10, 15, 20	5	—	25	Increased
80	25, 50, 100	9.67	—	—	—
81	27	22	—	—	—
82	50% marble+50% stone powder	49	8.66	—	—
92	10	20	—	—	No change
85	0, 5, 10, 10, 15, 20, 25, 30, 35, 40, 45, 50	20.7	19.2	—	—
75	0, 5, 7.5, 10, 15	22	15	Decreased	—
38	0, 25, 50, 100	10.72	45	13.13	Decreased
25	0, 20, 50, 100	19.2	18.18	—	—
(b)					
91	(5%–30%)15%	12	—	—	Decrease
88	(0%–20%) 10%	4.3	—	—	Decrease
89	0%–20%, 5%	8	4.8	—	Decrease
8	0%–20% (15%)	36	76	—	Decrease
11	0%–40% (40%)	34.9	82.14	Decrease	—
90	5%–30% (15%)	13.36	—	—	Decrease

coherence. These morphological and microstructural features collectively support the conclusion of strength enhancement due to the substitution of WMS (figure 6).

Furthermore, the interaction between tricalcium aluminate in cement and calcite in WMS enhances further the binding capability of the matrix while the carboaluminates formed due to dolomite and cement alkalis lead to a stronger matrix [13]. The addition of steel fibers greatly increased the f_{cts} by as high as an average of 15.1% for an addition of 1.5% steel fibres. In this respect, the concrete mixture containing 18% RHA, 36% WMS, and

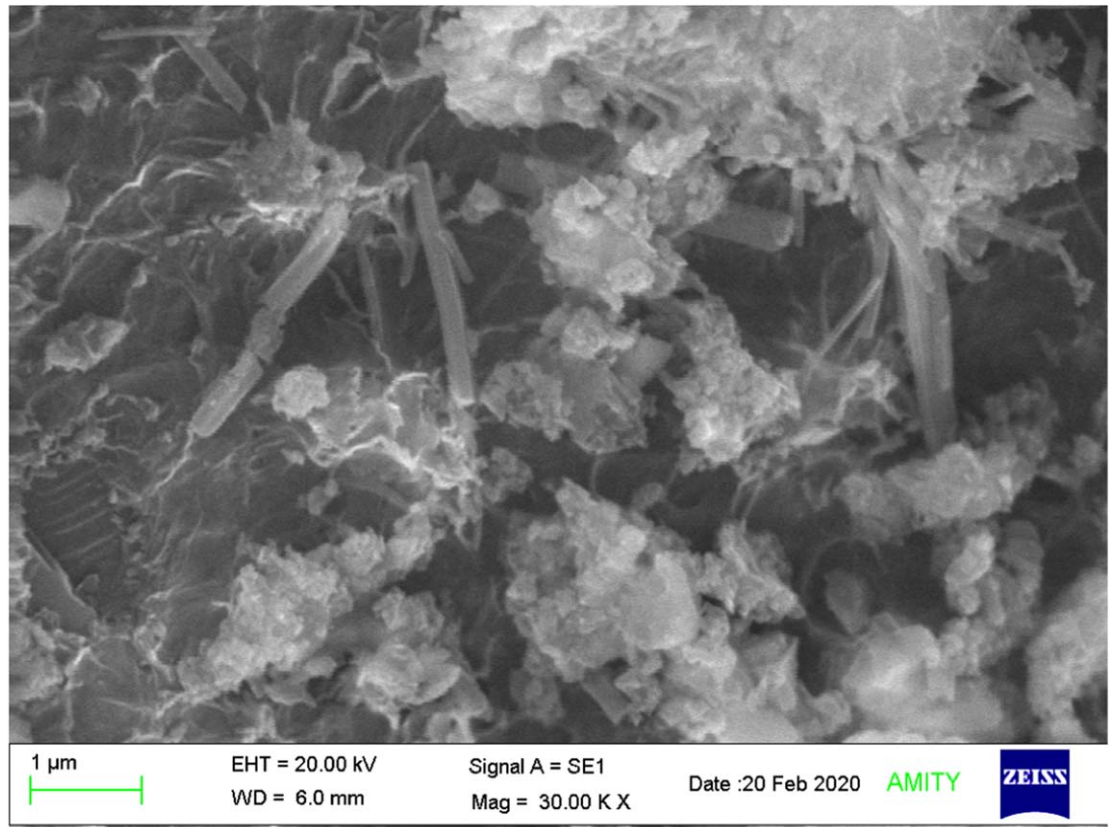


Figure 6. SEM image showing strength enhancement due to substitution of WMS.

1.5% steel fibres exhibited the highest increase in f_{cts} of **47.6%** when compared to the reference mixture (**RH0MD0SF0**). This is due to the fact that the fibres bridge the cracks through their bridging action, while good bonding between fibres and the cementitious matrix leads to an increase in f_{cts} and a decrease in the cracking of the matrix [24]. More elaborated results can be seen in the tables 7(a) and (b), showing that most researchers observed an improved mechanical performance with the addition of WMS and RHA. On the other hand, there are also some contradictory results reported in the literature pertaining to use of WMS in concrete[25]. The maximum enhancement of 47% in f_{cts} after steel fibres were added to lightweight concrete was recorded by Libre *et al* in 2011 [26]. Gao *et al* in 1997 [27] and Chen and Liu in 2004 [28] reported increases of 22% and 5%, respectively, while Hassanpour *et al* in 2014 reported an increase of 32% [29]. On the other hand, Campione *et al* 2005 determined no effect of steel fibres on f_{cts} [30], while some researchers like Duzgun *et al* 2005 [31] and El-Dieb *et al* 2009 [32] indicated very significant gains in f_{cts} due to steel fibre addition. Khaloo *et al* even reported reduced f_{cts} due to adding steel fibres [33]. The primary reasons relate to the positioning and orientation of steel fibres, which, to a large extent, accounts for the strength of concrete. Tables 7(a) and (b) show a relative summary of findings from previous research. The mix RH18MD36SF1.5 shows maximum enhancement of 47.6% in comparison to the reference mixture.

5. Split tensile strength (f_{spt}) results

f_{spt} was ascertained using cylindrical moulds (150 mm diameter and 300 mm height) as per IS: 5816 [34]. The cylindrical specimens were placed between the jaws of the universal testing machine, after which an axial load was applied along its diameter till occurrence of failure. The f_{spt} was obtained by dividing the maximum load by the cross-sectional area of the specimen. It can be noticed that from figures 7 and 8, the trend of f_{spt} is similar to f_{cts} peaking till 15% replacement of cement with RHA. From this figures it can be clearly seen that 28 day f_{spt} increased by **8.6%, 13.8%, and 17.1%** for **6%, 12%, 18%** substitution of cement with RHA respectively while f_{spt} decreases by 18.8 % for **24%** cement replacement with RHA. This trend is because of micro-filling effects leading to enhanced density of interfacial transition zone. Such trends have also been reported in previous literature [13]. Also, substitution of FAA with WMS increases the f_{spt} subsequently.

*In fact, at 28 days, the f_{spt} increased by **4.3%, 8.9%, and 16.3%** for **12%, 24%, and 36%** substitutions of FAA with WMS, respectively in comparison to the control mix. This is because of reduced pores and densified

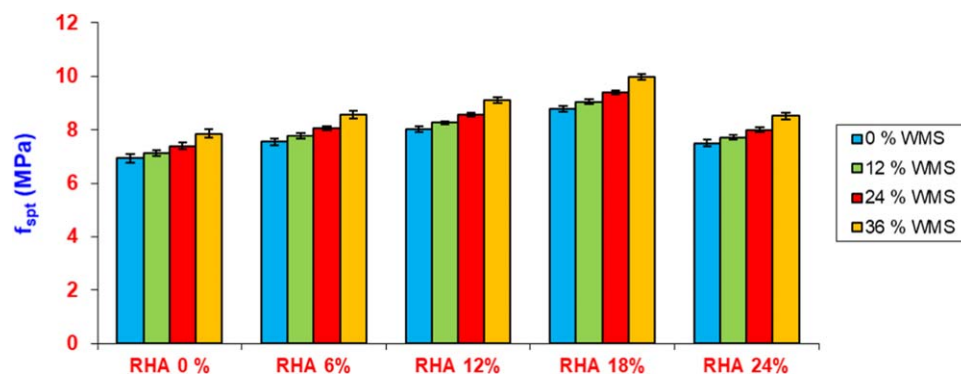


Figure 7. f_{spt} for concrete mixtures used in the experimental study without fibers.

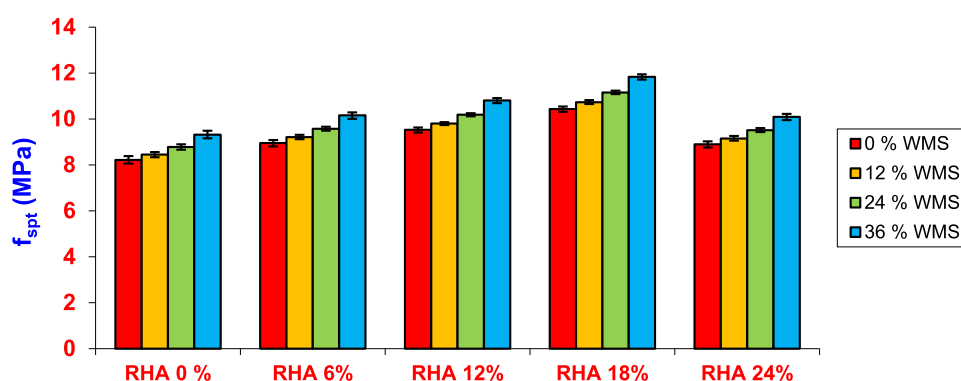


Figure 8. f_{spt} for concrete mixtures used in the experimental study with fibers.

cementitious matrix due to WMS inclusion [35]. Percentage improvement in tensile splitting strength is higher compared to the f_{spt} which could be attributed to the formation of carbo-aluminates. The experimental results agrees well with the earlier previous study as represented in tables 7(a) and (b). Hebhouh reported up to 33% gain in f_{spt} due to replacement of FAA with WMS [36] whereas Kabeer *et al* reported f_{spt} 23% increment in cement mortar cubes at 20% replacement of FAA with WMS [37]. Although, majority of the studies have reported positive impact of WMS inclusion in concrete with few exceptions like Sakalkale *et al* 2015 [38] and Silva *et al* [25] reported significant loss of mechanical properties due to inclusion of WMS at a replacement level of 50%. In the present reserach work, the addition of 1.5% HEF by weight of cement enhanced this property considerably, as depicted by an average increase in f_{spt} of about 22%. A increase of 45.8% for mixture RH18MD36SF1.5 in comparison to the reference mixture was observed. This improvement could be attributed to following reasons: (a) the steel fibres bridge the pre-existing cracks, hindering their further propagation, (b) the improvement in bond strength within the interfacial transition zone, and (c) the hooked ends of the fibres enhance the bond characteristic of the fibre matrix [39]. The improvement in tensile strength by fibre inclusion agrees with the earlier study. The maximum reported gain has been 116% by Libre *et al* (2011) for steel fiber-based composites [26]. Gao *et al* (1997) reported that f_{spt} increased proportionally higher compared to the f_{cts} [27]. Siddiqui (2016) reported a gain in the f_{spt} as high as 38.7% with 1.5% steel fibre content [39]. El-Dieb (2009) reported a gain of 52% due to addition of 0.52% steel fibres [32]. Sahmaran and Yaman 2007 [40] reported approximately 17% improvement in f_{spt} whereas Khaloo *et al* 2014 [33] and Gencel *et al* 2011 [41] reported a maximum increase of 28% and 21% in f_{spt} due to addition of steel fibres. Furthermore, Makorvic [42] developed the conclusion that the presence of short fibres acts as a catalyst that can develop higher tensile stresses than concrete containing with longer fibre only.

6. Flexural strength (f_{ft}) results

f_{ft} results were determined by casting beams of cross-sectional dimensions 150 mm $\times$ 150 mm with a span length of 700 mm. The f_{ft} results have been presented from figures 9, 10, which shows that the f_{ft} observes

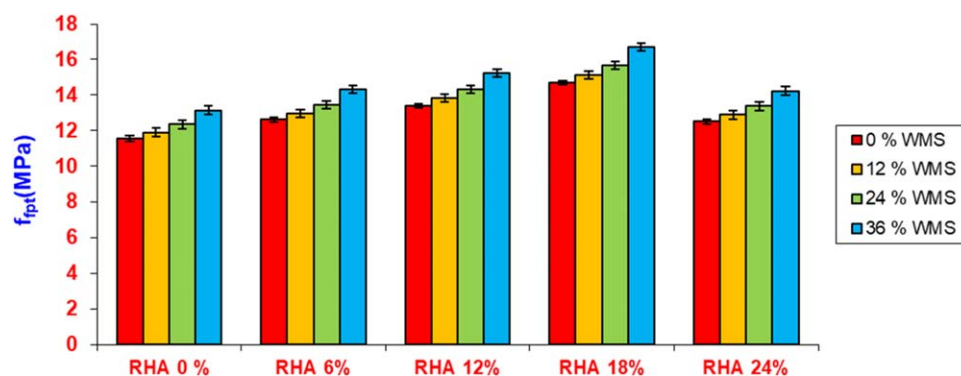


Figure 9. f_{cpt} for concrete mixtures used in the experimental study without fiber.

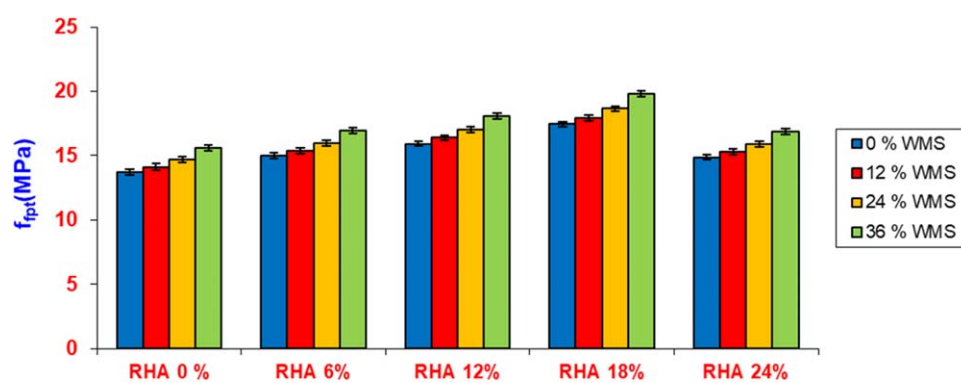


Figure 10. f_{cpt} for concrete mixtures used in the experimental study with fiber.

trend similar to f_{cts} , with the highest strength observed for the mix in which cement is substituted with 18% RHA. Also, 28 day f_{cpt} increased by **9.2%, 15.8%, and 17.6%** for **6%, 12%, 18%** substitution of cement with RHA respectively but f_{cpt} decreased by **8.7%** at 24% substitution of cement with RHA. It could be attributed to C-S-H provided by the RHA in addition to enhanced density of the cementitious matrix. The replacement of FAA by WMS also led to increase in f_{cpt} , as elucidated in figures 9 and 10. In fact, at 28 days, the f_{cpt} increased by **4.2%, 9.0%, and 15.5%** for **12%, 24%, and 36%** substitutions of FAA with WMS, respectively in comparison to the reference mixture. This is due to higher density in the cementitious matrix, and improved binding properties due to WMS addition [41]. Further, the carbo-aluminates due to reaction of tricalcium aluminate with calcite increases the f_{cpt} subsequently [13]. Similar trend has been observed as reported in previous studies in table 9. Similarly, researchers like Sakalakale *et al* 2014 [38] and Rai *et al* 2022 [43] observed enhancement in f_{cpt} by 10 % due to FAA substitution with WMS. In the present reserach work, incorporation of HEF significantly raises the f_{cpt} , with an average increase of 35% with the addition of 1.5% HEF. In fact, the RH18MD36SF1.5 mixture observed a maximum increase in f_{cpt} of about **49.6%** over the reference mixture, RH0MD0SF0. This could be attributed to high aspect ratio and tensile strength of the fibres. Moreover, hooked steel fibres improve the fiber-matrix bond and release fracture energy at crack tips which allows the tensile stresses to be effectively transferred to the fibres, thereby preventing the propagation of macro-cracks [13]. Also, higher f_{cpt} due to the random distribution of the fibres, which restricted crack propagation leading to higher f_{cpt} . These observations fell in line with Khaloo *et al* (2014) who observed a 46% increase in f_{cpt} [33]. Likewise, Oliveira *et al* (2013) found that inclusion of steel fibres from 0.5% to 1%, resulted in a 100% increase in f_{cpt} , but when the steel fibre percentage was further increased to 1.5%, this rate of increase reduced to only a 25 % [44]. Tables 7(a) and (b) provide comparisons with the test results of previous studies.



Figure 11. Workability of the concrete mixtures used in the experimental study without fiber [Series 1 denote 0% FAA substitution by WMS; Series 2 denote 12% FAA substitution by WMS; Series 3 denote 24% substitution by WMS; Series 4 denote 36% substitution by WMS].

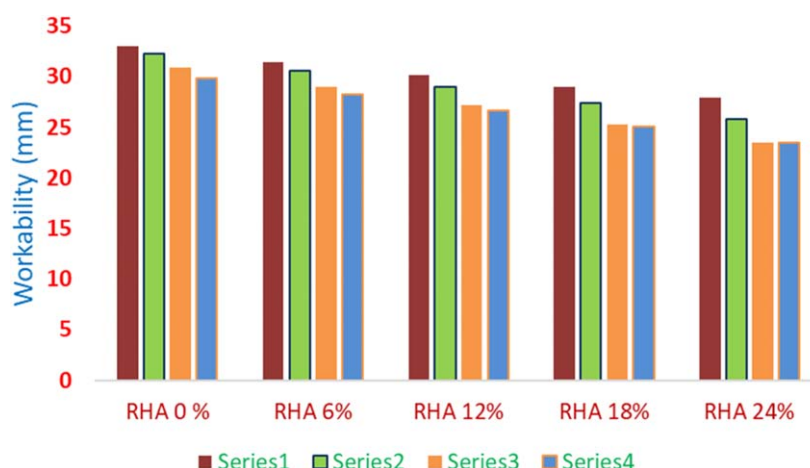
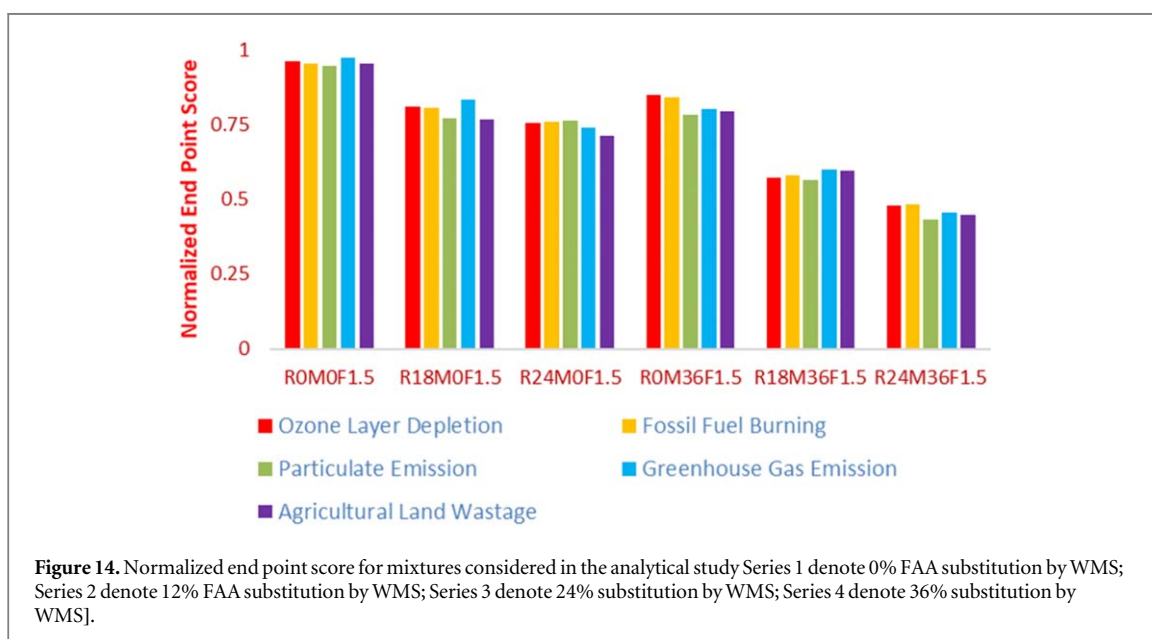
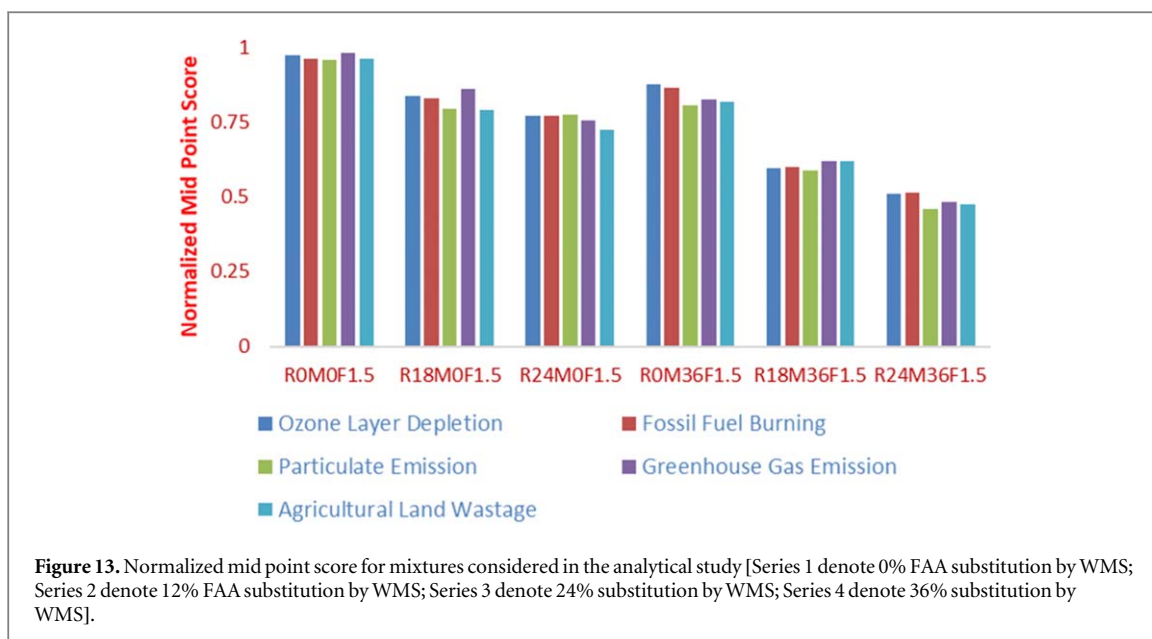


Figure 12. Workability of the concrete mixtures used in the experimental study with fiber [Series 1 denote 0% FAA substitution by WMS; Series 2 denote 12% FAA substitution by WMS; Series 3 denote 24% substitution by WMS; Series 4 denote 36% substitution by WMS].

7. Concrete slump test

The slump test (to determine workability) was conducted as per procedure described in BIS 1199–1959 [45]. The slump test, conducted as per BIS 1199–1959, involves filling a standard slump cone (300 mm height, 200 mm base diameter, 100 mm top diameter) in three layers, each compacted with 25 strokes of a tamping rod. The cone is lifted vertically, and the subsidence of the concrete is measured as the slump, indicating workability. The measured slump provides a quick assessment of the concrete's consistency and fluidity.

The resulting slump values are presented in figures 11 and 12, which indicate that workability decreased by 16.1% when 18% of cement was replaced by RHA. This is because, as the increased surface area of the binder when RHA content rises, necessities requirement of more water to maintain proper consistency [7]. According to Kang *et al* (2020), the use of RHA results in higher water absorption, thereby reducing workability [46] The workability reduces by 9.8% when FAA is replaced by 36% WMS. Additionally, since WMS has a larger surface area than river sand, a higher amount of water is needed to wet the surface, significantly reducing workability. Workability also decreased with the addition of fibres by 7.4%. This is because HEF resulted in congestion within the concrete mix, impeding its flow. The extent of this congestion is proportional to the HEF content. Previous researchers also noted a decrease in workability with increased HEF content [7, 13, 47].



8. Environmental impact assessment (EIAA)

EIAA is essential for sustainability because it helps identify, evaluate, and mitigate potential environmental risks associated with various activities, ensuring that resources are used responsibly, and ecosystems are protected. A sustainable approach requires minimizing harmful emissions, reducing waste, and conserving natural resources to meet the needs of the present without compromising the future. By quantifying environmental impacts, stakeholders can make informed decisions, adopt eco-friendly technologies, and foster a circular economy, aligning growth with environmental conservation (figures 13 and 14).

Replacing cement with RHA and FAA with WMS addresses SDG 11 (Sustainable Cities and Communities) by promoting sustainable construction materials. Also, it is in alignment with SDG 12 (Responsible Consumption and Production) by reducing the demand for natural resources and diverting agricultural and industrial waste from landfills. Additionally, these material substitutions contribute to SDG 13 (Climate Action) by reducing the carbon footprint associated with cement production, which is a major emitter of CO₂. The reuse of waste materials also promotes a circular economy, helping to achieve a more sustainable construction sector. In last few years tremendous work has been environmental impact assessment (EIAA) with Klemeš *et al* (1997) developing a novel approach for reducing carbon dioxide emissions resulting from fossil fuel combustion [48]. Later, Klemeš *et al* (2007) conducted techno-economic modelling to ascertain CO₂ emissions and process

Table 8. Comparison of different Environmental impact assessment methods (EIA).

Aspect	EIA methods	ReCiPe midpoint	ReCiPe endpoint	Advantages	Disadvantages	Applicability
Scope	Project or policy-specific	Product or process-specific (LCA)	Aggregated, holistic overview (LCA)	Regulatory compliance; Highlights project-specific issues	Limited lifecycle perspective; Does not aggregate across broad impacts	Environmental project assessments (e.g., construction, infrastructure)
Type of Assessment	Qualitative and spatial	Quantitative, focusing on specific indicators	Quantitative, aggregated into broad impacts	Clear visual tools (e.g., matrices, maps); Cost-effective for policy assessments	Subjective results; Can miss cumulative impacts	Regional policies, compliance reporting
Level of Detail	Varies; some are qualitative, some quantitative	Highly detailed with 18 indicators	Aggregated into 3 categories	Adaptable for small or large-scale projects	Requires significant expertise for large projects	Large-scale development projects
Purpose	Evaluate project feasibility and compliance	Measure environmental impacts across product life cycles	Provide broader view of impact on health, ecosystems, and resources	Ensures project sustainability and regulatory adherence	Not ideal for complex, cumulative impact assessments	Sustainability assessments for infrastructure
Indicator Focus	Depends on project impacts (e.g., air, water)	Specific environmental issues like climate change	Human health, ecosystem quality, resource depletion	Flexible and tailored to the specific project context	Not suitable for product-level comparisons	Project-based impact analysis
Temporal Scope	Snapshot of project's life	Full life-cycle perspective	Full life-cycle perspective	Easy to implement; No need for life-cycle data	Misses long-term and global environmental impacts	Short-term impact assessments
Data Requirements	Can use qualitative and expert opinion	Requires detailed input data for the LCA model	Similar to midpoint but aggregated	Requires less data than LCA methods	Less precise than LCA approaches	Regulatory EIA compliance
Example Application	Assessing environmental impact of a new road project	Calculating carbon footprint of a product	Evaluating the overall environmental burden of a product lifecycle	Applicable to construction, policy, and regional projects	Less effective for product or process comparisons	Infrastructure and development projects

Table 9. Inventory data available for concrete (1 Cubic Meter) [Varadharajan et al. 2020, Reference 7, 13].

Raw material	Quantity	Units
Cement	475	kg
Sand	610	kg
Crushed Gravel	1100	kg
Water	205	kg
Electricity	25.47	kWH
Diesel	20.88	MJ
Cooling Water	92.7	kg
Explosives	0.02124	kg
Steels	3.807	kg
Silica Sand	1.089	kg
Abrasives	0.08532	kg
Timber	1.161	kg
Coagulant and Flocculents	0.0189	kg
Oil and Grease	0.03654	kg
Carbon Dioxide	0.0972	kg
Lime Stone	128.8	kg

integration techniques have been recently used [49]. Also, process integration have often been used for EIAA [13]. In calculating environmental impacts with respect to the principles of sustainable development, energy analysis was used by Giannetti *et al* in 2006 [50]. Sevegnani *et al*, in 2011, extended this tool to evaluate city sustainability according to renewable energy criteria [51]. Further, Mikulčić *et al* reported beneficial use of alternatives to replace cement in decreasing CO₂ based on Life cycle assessment (LCA) study [52]. LCA of concrete was carried out for the present research work based on the ISO 14040 [53]. The different phases of LCA as laid down by ISO 14044(2006) include: (a) Defining scope and Goal, (b) Analysis of Inventory, and (c) EIAA. Comparison of adopted method with different environmental impact assessment methods have been shown in table 8.

8.1. Defining scope and goal

The main aim was to conduct EIAA due to use of RHA & WMS for replacement of Cement and FAA. OPEN LCA Software [54] was used for conducting EIAA. The study involves an LCA of the concrete stages from production to disposal. The concrete has been measured in cubic meters (m³); the percentage for replacement of cement by RHA varies from 0% to 24%, and substitution of FAA by WMS will vary from 0% to 36%.

8.2. Analysis of inventory

The software used for LCA analysis in the present research work is OPEN LCA and the database used for the study is Eco-Convenient Database. To conduct LCA analysis, the energy input is taken from Mendoza *et al* (2014) [55] and other relevant data is taken from Varadharajan *et al* 2020 [7, 13] as shown in table 9. Recipe midpoint and endpoint analysis has been used for conducting EIAA.

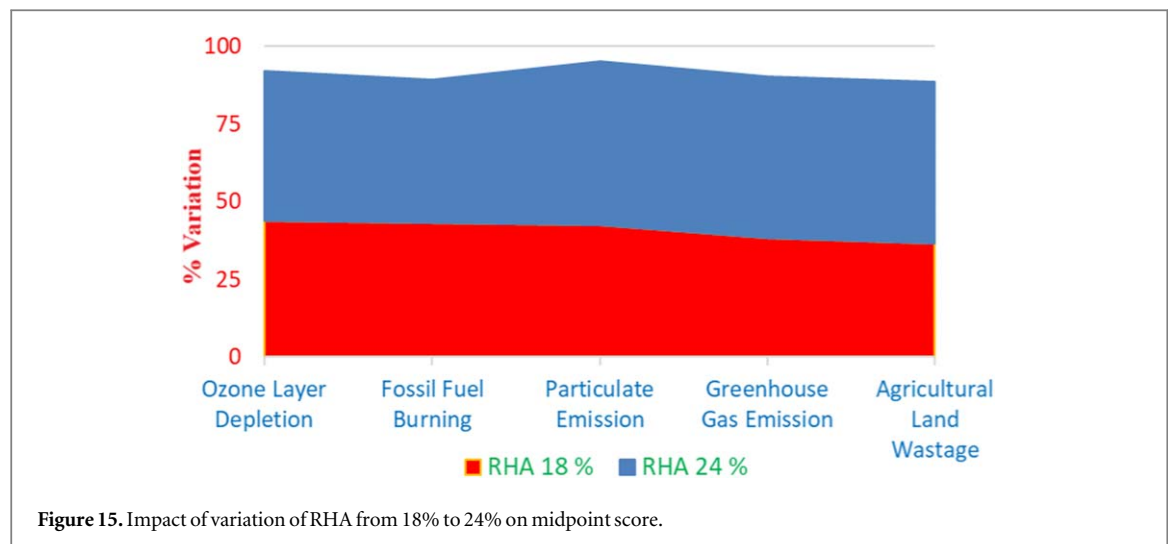
8.3. EIAA

The EIAA model used for all mixtures has been kept constant, while the environmental impact analysis was varied by adjusting the quantities of RHA and WMS in the concrete. This study encompasses all stages of concrete, from production to disposal.

8.3.1. Mid point assessment

The environmental impact assessment shown in the graph highlights the advantages of using RHA as a partial cement replacement and WMS as a partial FAA substitute in concrete. The parameters discussed in the study are greenhouse gas emissions, fossil fuel consumption, particulate emissions, ozone layer depletion, and agricultural land wastage. In this study the environmental impact assessment of best mix will be presented since the main aim of the study is to identify a mix with maximum mechanical properties with substantial environmental benefits.

Replacing 18% cement with RHA & FAA with 26 % WMS leads to a significant reduction in the carbon footprint of concrete by around 37.4%. RHA is an agricultural byproduct that requires minimal processing and emits less CO₂, even when accounting for transportation. The energy savings from reduced cement production also directly impact fossil fuel consumption, as energy-intensive clinker production accounts for a significant



share of the cement industry's energy demand. This aligns with UN SDG 13 (Climate Action) by reducing emissions and promoting low-carbon technologies.

Concrete production involves substantial water usage, both in cement manufacturing and in mixing processes. The partial substitution of cement with RHA can reduce the overall water footprint of the construction process. Cement production requires approximately 100–150 liters of water per ton for cooling and dust suppression. Since RHA is a byproduct that does not require intensive processing, the water consumption in the concrete mix decreases by 14.33%, contributing to SDG 6 by advocating efficient use of water resources.

The use of WMS as FAA substitute addresses the critical issue of resource depletion. Quarrying of sand and aggregates not only depletes finite natural resources but also leads to habitat destruction and environmental degradation. Substituting 36% WMS in place of FAA & 18 % RHA for cement replacement reduces the dependence on river sand, crushed stone and other natural resources, contributing to UN SDG 12 (Responsible Consumption and Production) by fostering resource conservation and minimizing environmental damage as indicated by reduction in fossil fuel burning (39.4%). Additionally, WMS is an industrial byproduct that would otherwise end up in landfills, contributing to solid waste generation. The incorporation of 36 % WMS & 18 % RHA in concrete supports waste diversion from landfills, aligning with the principles of a circular economy and promoting industrial symbiosis.

The incorporation of 36 % WMS and 18 % RHA also has a positive impact on air quality by mitigating dust and particulate emissions (40.9%). The reduction in natural aggregate processing, which is often associated with significant dust generation, contributes to improved air quality. Furthermore, the avoidance of rice husk burning, which is common in agricultural waste disposal, prevents the release of fine particulate matter (PM_{2.5} and PM₁₀) and volatile organic compounds (VOCs), thereby contributing to SDG 3 (Good Health and Well-being) by reducing air pollution and improving public health outcomes.

The utilization of 18 % RHA & 36 % WMS promotes sustainable agricultural waste management, reducing the environmental burden of open-field burning, which is a common practice in rice-producing regions. By transforming rice husk waste into a valuable construction material, the need for large landfill areas is minimized, contributing to agricultural land conservation (36.4%). This directly addresses UN SDG 15 (Life on Land) by promoting sustainable waste reuse practices that reduce environmental degradation.

Although the graph shows modest improvements in ozone layer depletion, the use of alternative materials reduces emissions of NO_x and SO_x gases generated during cement production, which are known to contribute to ozone depletion. This aligns with SDG 11 (Sustainable Cities and Communities) by fostering cleaner industrial practices and protecting the environment from harmful emissions.

In summary, the combination of 18% RHA as cement substitute and WMS as substitute of FAA (36%) in concrete results in a comprehensive benefits. These improvements align with several UN SDGs, like 3,6,11,12,13 & 15. The adoption of RHA and WMS not only contributes to a circular economy but also promotes eco-friendly construction practices, paving the way for a more sustainable future in the building sector (figures 15–17).

8.3.2. End point assessment (EPA)

EPA shown in the graph captures the environmental impact of concrete from the placement stage through its lifecycle to disposal, incorporating factors such as greenhouse gas emissions, fossil fuel consumption, particulate emissions, ozone layer depletion, and agricultural land wastage. The normalized endpoint scores demonstrate

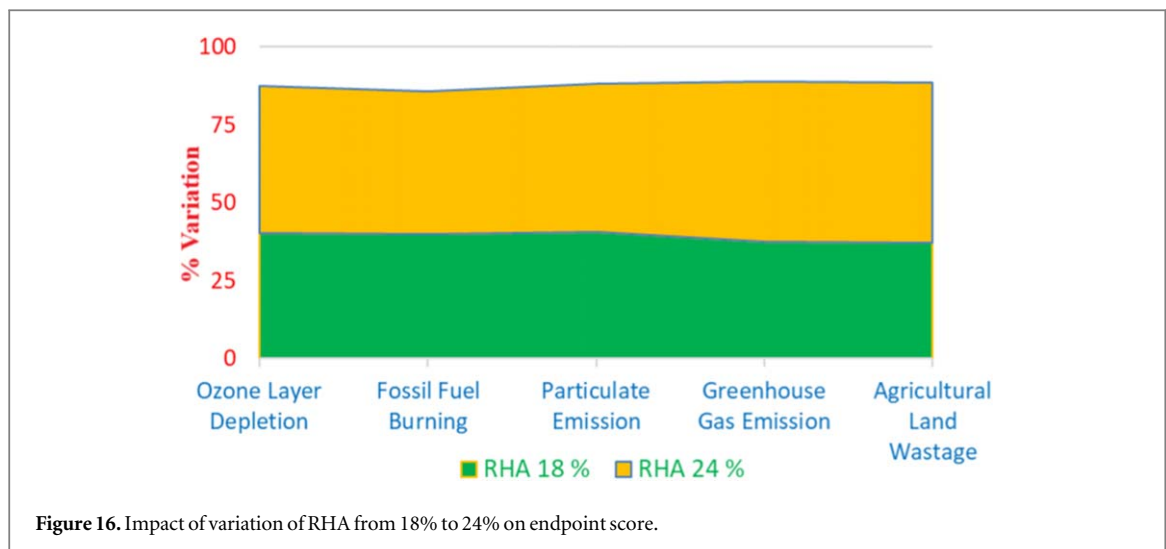


Figure 16. Impact of variation of RHA from 18% to 24% on endpoint score.

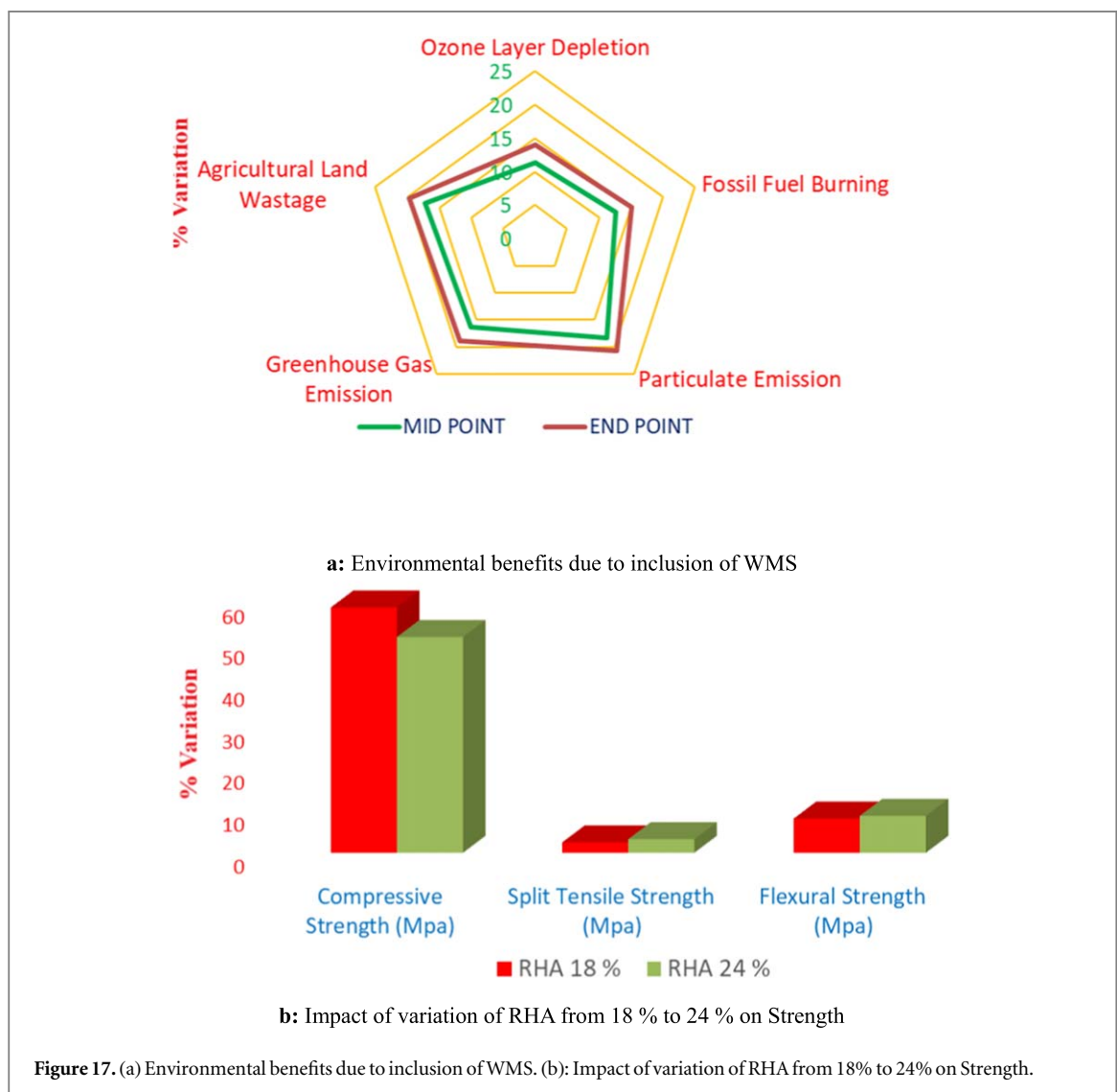


Figure 17. (a) Environmental benefits due to inclusion of WMS. (b): Impact of variation of RHA from 18% to 24% on Strength.

that increasing the proportion of rice husk ash (RHA) and waste marble powder (WMS) significantly reduces environmental burdens, reflecting the sustainable performance of these concrete mixes throughout their lifecycle.

The reduction in fossil fuel consumption, indicated by lower scores with increasing RHA and WMS percentages, is a direct result of reduced demand for energy-intensive cement and natural aggregate production

(42.1%). This reduction aligns with UN SDG 12 (Responsible Consumption and Production) by minimizing the use of non-renewable energy resources, promoting sustainable material consumption, and encouraging circular economy practices. The mix with 18% RHA and 36% WMS (RH18MD36SF1.5) shows the most significant reduction in agricultural land wastage, highlighting the role of agricultural and industrial waste reuse across the concrete lifecycle (39.1%).

The endpoint assessment also shows a clear decline in greenhouse gas emissions as the percentage of RHA & WMS increases, with the RH18MD36SF1.5 mix demonstrating the lowest emissions. Cement manufacturing is a major contributor to CO₂ emissions globally, and substituting cement with RHA can reduce these emissions by up to 38.6% over the concrete's lifecycle, including disposal. This aligns with UN SDG 13 (Climate Action) by supporting efforts to mitigate climate change through carbon reduction strategies.

The scores for particulate emissions decline as well, particularly in mixes with higher WMS & RHA content. Using WMS in place of FAA reduces the need for quarrying and mining along with burning of rice husk, processes that contribute to air pollution and particulate matter emissions (43.2%). This outcome supports UN SDG 3 (Good Health and Well-being) by reducing pollutants that harm respiratory health, especially in urban and industrial areas.

The assessment indicates a decline in agricultural land wastage when RHA is used. Incorporating rice husk ash ensures that agricultural byproducts are utilized effectively, reducing the need for open-field burning and landfilling. This practice supports UN SDG 15 (Life on Land) by preventing land degradation and promoting sustainable waste management practices. Furthermore, the reuse of waste materials like WMS reduces landfill dependency, preventing land wastage and aligning with sustainable urban development goals.

In terms of ozone layer depletion, the reduction, although moderate, reflects the decreased emissions of NO_x and SO_x gases associated with cement production. Substituting cement with RHA and aggregates with WMS contributes to reduction in ozone depletion (37.6%) over the concrete's lifecycle, which aligns with UN SDG 11 (Sustainable Cities and Communities) by promoting cleaner construction practices and minimizing pollution.

The endpoint assessment shows that replacing 18% of cement with RHA and 36% FAA with WMS without steel fibers delivers substantial environmental benefits across the lifecycle of concrete. These substitutions reduce fossil fuel consumption, greenhouse gas emissions, particulate emissions, and land wastage, all of which contribute to achieving multiple UN SDGs. Specifically, these practices align with SDG 3 (Good Health and Well-being), SDG 11 (Sustainable Cities and Communities), SDG 12 (Responsible Consumption and Production), SDG 13 (Climate Action), and SDG 15 (Life on Land). By integrating agricultural and industrial waste into concrete, these mixes promote sustainable waste management, reduce environmental degradation, and advance the transition toward a low-carbon, circular economy. This demonstrates how sustainable material choices can optimize environmental performance throughout the entire lifecycle of construction materials.

The environmental impact of all the processes involved in concrete production will be judged by midpoint analysis. Figures 15–17 shows that the replacement of cement by RHA and FAA by WMS has a positive impact on the environment. Greenhouse gas emissions and particulate emissions are two important factors for environmental impact assessments. These are reduced by 36.6% and 36.5% with the use of 36 % WMS and 18 % RHA with 0% Steel fibres, respectively. In the past three decades, ozone layer depletion has been one of the primary causes of concern. The addition of RHA and WMS reduces the depletion of the ozone layer by 34.8%. Setting up concrete involves massive amount of burning of fossil fuel in order to produce and process its raw materials. Substitution by WMS and RHA reduces fossil fuel burning significantly by 41.7%. Furthermore, as earlier stated, careless disposal of RHA and WMS clogs up the soil's pores, making it barren or incapable of being arable, consequently impacting agricultural land use. Aquifers are degraded by reducing its capacity due to no infiltration of rain water. EIAA shows that incorporation of RHA & WMS increase agricultural land availability by 37.2% , which is quite valuable especially for countries with agriculture based economy. The individual impacts of WMS and RHA, respectively, are elaborated further in figures 13–17. Even though, cement replacement by RHA yields considerable environmental benefits but it substantially reduces the strength of concrete (figure 17). Therefore mix RH18MD36SF1.5 is the optimum mix yielding both strength and environmental benefits.

9. Cutting-edge adaptability in construction: optimizing practices for enhanced performance using rice husk ash and marble waste powder in concrete

Incorporating alternative materials like rice husk ash (RHA) as a cement replacement and waste marble slurry (WMS) as a fine aggregate replacement in concrete offers a sustainable approach to modern construction challenges. Waste marble slurry, a byproduct of marble processing, is rich in calcium carbonate and has great potential as a micro-filler in concrete. However, adopting RHA and WMS requires adjustments to construction

practices to ensure compatibility with international building standards. This section provides a comprehensive technical analysis of necessary construction practice adjustments, backed by microstructural insights and considerations for code compliance.

9.1. Microstructural analysis and strength mechanisms

RHA's high amorphous silica content makes it a valuable pozzolanic material, reacting with calcium hydroxide (CH) generated during cement hydration to form additional calcium silicate hydrate (C-S-H) gel, which strengthens and densifies the concrete matrix.

- **Pozzolanic Reaction and C-S-H Gel Formation:** The reaction of amorphous silica in RHA with CH results in additional C-S-H gel, which significantly enhances the microstructure by filling in voids, refining pores, and improving the interfacial transition zone (ITZ) between the cement paste and aggregates. This improved ITZ leads to enhanced tensile and compressive strength by reducing weak zones within the concrete matrix. The interfacial transition zone (ITZ) in concrete is the weak region between cement paste and aggregates, characterized by higher porosity and microcracks. It significantly influences the mechanical properties and durability of concrete.
- **Reduction in CH Content:** By consuming CH, which is typically a weak component in traditional concrete, RHA effectively replaces it with more durable C-S-H gel. This reduces vulnerability to chemical attack, particularly from sulfates and chlorides, which aligns with durability standards in codes like ACI 318 [56] and Eurocode 2 [57].
- **Finer Microstructure with Reduced Permeability:** The fine particles in RHA act as micro-fillers, sealing voids in the matrix. This pore refinement contributes to lower permeability, which is crucial for durability, especially in structures exposed to water or aggressive ions, such as marine environments. This characteristic aligns with the permeability standards outlined in ASTM C1202 [58].
Waste marble slurry, composed predominantly of calcium carbonate, serves as a fine aggregate replacement that enhances the concrete's density and durability through its micro-filler properties.
- **Filler Effect and Improved Packing Density:** WMS particles, being finer than traditional sand, effectively fill micro-voids in the concrete matrix, resulting in a densely packed structure. This reduces the total void content and improves the packing density, which reduces permeability and increases strength, aligning with guidelines in ISO 22965 [59] for concrete density.
- **Calcium Carbonate Reaction Potential:** The calcium carbonate in WMS can engage in mild carbonation reactions that further stabilize the matrix. Additionally, when used alongside RHA, WMS may facilitate secondary pozzolanic reactions, creating additional C-S-H gel and enhancing the ITZ.
- **Thermal and Shrinkage Behavior:** WMS improves thermal stability and reduces shrinkage, as the calcium carbonate in WMS can help to mitigate cracking due to thermal expansion and contraction. This aligns with ASTM C157 [60] standards, especially beneficial in regions with fluctuating temperatures.

9.2. Required adjustments in construction practices

To maximize the benefits of RHA and WMS and ensure compatibility with international standards, several adjustments to construction practices are necessary.

9.2.1. Revised mix design procedures

Concrete mix design must be adjusted to account for the unique properties of RHA and WMS, following protocols in ASTM C495 [61] for blended cements.

- **Optimized Proportioning:** Generally, replacement levels range from 10–20% for RHA as a cement replacement and 10–30% for WMS as a fine aggregate replacement. These percentages should be fine-tuned based on specific project requirements and validated to ensure compliance with standards such as ACI 211.1 [62] and EN 197-1 [63].
- **Water-Cement Ratio and Admixture Use:** RHA can reduce the water demand due to its fineness and reactivity, while WMS may slightly increase water demand due to its finer particle size. Therefore, high-range water reducers (superplasticizers) may be necessary to maintain workability without increasing the water-cement ratio, aligning with ASTM C494 [64] specifications.

9.2.2. Enhanced curing practices

9.2.2.1. Optimal curing conditions are essential to fully realize the benefits of RHA and WMS.

- **Extended Moist Curing:** Extended moist curing of 28 days or longer is recommended for concrete with RHA to allow the full development of pozzolanic reactions. This extended curing is especially important for structures requiring enhanced durability, as specified by ASTM C511 [65].
- **Controlled Curing for Shrinkage Reduction:** RHA and WMS both contribute to reduced shrinkage; however, controlled curing is still recommended to avoid premature drying, especially in hot climates. This measure supports durability and is consistent with Eurocode 2 [66] recommendations for shrinkage control in concrete.

9.2.3. Precision in batching and mixing

Given the fine particle size of RHA and WMS, precision in batching and mixing is crucial to maintain consistency and achieve uniform performance.

- **Extended Mixing Time:** Following ASTM C94 [67], slightly extended mixing times are recommended to ensure thorough blending of RHA and WMS and to avoid segregation. Proper mixing ensures consistent strength across the structure by minimizing areas of low particle distribution.
- **Automated Dosing for Consistency:** Automated dosing equipment is recommended to maintain precise control over the proportions of RHA and WMS, ensuring compliance with ACI 301 [68] quality standards and enhancing uniformity in the final concrete product.

9.2.4. On-site workability adjustments and compaction

Due to the finer particle sizes of RHA and WMS, adjustments are necessary for workability and compaction to meet standards like ISO 22965:2021 [69].

- **Use of Superplasticizers:** High-range water reducers are suggested to improve workability without raising the water-cement ratio. This is particularly important in applications with high RHA content where reduced workability may affect placement and compaction efficiency.
- **Compaction Techniques:** Proper compaction is necessary to achieve maximum density in the RHA-WMS concrete mix. Mechanical vibrators should be used according to ASTM C31 [70] guidelines to eliminate air voids, enhancing strength and reducing permeability.

9.2.5. Quality control and compliance testing

Enhanced quality control is crucial to ensure that RHA-WMS concrete meets the performance standards specified in international building codes.

- **Strength Testing at Extended Intervals:** Given the ongoing pozzolanic activity of RHA, compressive and flexural strength tests should be conducted at intervals beyond the typical 28-day period, such as at 56 days. This provides a comprehensive assessment of long-term performance and ensures compliance with ACI 318 [56] and EN 12390-3 [71] standards.
- **Durability Testing for Chloride and Sulfate Resistance:** Durability tests, including ASTM C1202 [58] for chloride ion penetration and ASTM C1012 [72] for sulfate resistance, are recommended to confirm that the reduced permeability and increased resistance to chemical attacks in RHA-WMS concrete meet durability standards.

9.3. Microstructural rationale for proposed practices

9.3.1. Each of the recommended construction adjustments is grounded in the microstructural behavior of RHA and WMS:

- **Enhanced ITZ Strength:** RHA densifies the ITZ, reducing weak points and enhancing bonding with aggregates. WMS further strengthens this zone through improved packing, aligning with durability and mechanical strength requirements specified in international standards.

- **Refined Pore Structure:** RHA's fine particles reduce pore size, while WMS fills voids and increases packing density. This microstructural refinement minimizes permeability, which is crucial for chloride and sulfate resistance, as required by standards like ASTM C1202 [58].
- **Thermal Stability and Reduced Shrinkage:** WMS's calcium carbonate composition provides thermal stability and reduces shrinkage. This behavior is beneficial for structures in climates with temperature fluctuations, aligning with ASTM C157 [60] for dimensional stability.

10. Special construction practices with reinforcement detailing impact for structural members like beams, columns and slabs

For beams, which primarily experience bending stresses, incorporating RHA as a cement replacement (0–24%) and WMS as a fine aggregate replacement (0–36%) enhances both compressive and tensile performance. The pozzolanic activity of RHA generates additional C-S-H gel, which densifies the concrete matrix, particularly around the reinforcement bars. This densification improves the bond between concrete and steel, which is critical for beams where load transfer relies on this bond strength. The fine particles in WMS fill micro-voids [13, 73, 74], reducing porosity and minimizing pathways for moisture and aggressive ions, leading to enhanced corrosion resistance for embedded steel reinforcement. This is a significant advantage for beams exposed to varying environmental conditions.

In terms of reinforcement detailing, the refined Interfacial Transition Zone (ITZ) created by RHA and WMS contributes to stronger bonding with rebar, effectively anchoring the steel within the concrete matrix. This improved anchorage reduces the risk of slippage or bond failure, even under high tensile loads. Additionally, the dense microstructure minimizes crack widths around the rebar, distributing tensile stresses more effectively along the length of the reinforcement. By reducing micro-cracks and enhancing the concrete's durability, RHA and WMS concrete ensures that beams maintain their integrity under load, extending their service life and reducing maintenance needs.

For slabs, which require resistance to both distributed loads and environmental exposure, the use of RHA (0–24%) and WMS (0–36%) helps in achieving a compact, impermeable concrete surface. The addition of RHA improves the slab's resistance to surface cracking and abrasion, while WMS acts as a filler, reducing porosity and enhancing the slab's durability against moisture ingress. This is particularly beneficial for exposed slabs, such as pavements or slabs in high-humidity environments, as the dense microstructure resists weathering and chloride penetration.

In terms of reinforcement detailing, the enhanced ITZ provided by RHA and WMS supports a stronger bond between concrete and reinforcement, crucial for flexural strength in slabs. The refined microstructure minimizes the formation of cracks around the rebar, which is vital for durability and helps distribute tensile forces effectively. Additionally, the reduced permeability around reinforcement reduces corrosion risks, even in coastal or industrial environments where chloride exposure is a concern. This improved durability and reduced risk of reinforcement corrosion in slabs help maintain structural integrity over time, ensuring load-bearing capacity and minimizing long-term maintenance costs.

Columns, being primary load-bearing elements, benefit from the compressive strength enhancement provided by RHA (0–24% replacement) and WMS (0–36% replacement). RHA's pozzolanic activity contributes additional C-S-H gel, which increases the compressive strength of the concrete. WMS fills voids within the column's concrete matrix, resulting in reduced porosity and increased density, which is essential for columns under axial loads. These modifications enhance the load-bearing capacity and stability of columns, ensuring they meet performance standards for structural resilience.

Reinforcement detailing in columns is critical for confinement, and the use of RHA and WMS plays a vital role in improving this. The dense concrete matrix around the rebar improves bonding and enhances the load transfer between concrete and steel. This enhanced bond strength reduces potential slippage, improving the confinement of rebar under load and preventing buckling of longitudinal reinforcement. Additionally, the reduced permeability provided by RHA and WMS reduces the risk of steel corrosion within the columns, which is crucial for long-term durability in aggressive environments. By minimizing corrosion and strengthening the bond with reinforcement, RHA and WMS concrete contribute to the stability and longevity of columns, making them more resilient to axial and lateral loads over their lifecycle. The summary of recommendations are described in table 10.

Table 10. Impact of using RHA and WMS on different stages of construction.

Aspect	Impact of RHA	Impact of WMS
Mix Design	RHA requires careful water-cement ratio adjustments due to its fineness.	WMS increases the overall particle packing, leading to reduced voids and higher density.
Construction	Extended mixing time is needed to ensure even distribution and pozzolanic reactivity.	Finer particles can lead to segregation; precise batching and mixing are necessary.
Curing	RHA requires extended moist curing (28–56 days) to fully activate pozzolanic reactions.	Standard curing practices are usually sufficient, but WMS benefits from moist curing to enhance matrix bonding.
Concrete Placing	Additional attention to compaction to prevent segregation and ensure proper placement.	WMS's fine particles aid in achieving a smooth, consistent mix but require careful handling to avoid segregation.
Finishing	RHA increases density, resulting in a smooth finish that resists surface abrasion.	WMS contributes to a refined surface finish with fewer surface voids and improved durability.
Maintenance	RHA reduces permeability, lowering maintenance due to less water penetration and cracking.	Dense concrete with WMS minimizes surface erosion and is more resistant to weathering.

11. Conclusions

The findings from this study underscore the remarkable potential of integrating RHA and WMS into concrete, yielding improvements in both mechanical properties and environmental sustainability. Substituting cement with RHA up to a 18% replacement level resulted in notable enhancements in compressive, split tensile, and flexural strengths. This strength improvement is primarily due to the pozzolanic activity of RHA, which promotes the formation of additional calcium silicate hydrate (C-S-H) during secondary hydration. This reaction densifies the concrete matrix, filling microvoids and enhancing bond strength between particles. However, when RHA replacement increased to 24%, a decline in strength was observed, likely due to the reduction in cement content, which restricted the hydration process and diminished the availability of primary binding agents. Thus, the optimal RHA replacement was identified as 18%, balancing strength improvement with cement reduction.

Similarly, replacing fine aggregate (FAA) with WMS up to 36% considerably boosted concrete's mechanical properties, with the 36% WMS mix achieving peak values in compressive, split tensile, and flexural strengths. This performance enhancement is attributed to the filler effect of finely powdered WMS, which increases particle packing density, decreases void spaces, and reduces porosity in the concrete matrix. Additionally, the slight cementitious nature of WMS further contributes to strength gains by creating secondary hydration bonds. However, with WMS replacements exceeding 36%, challenges like segregation and reduced cohesion may arise. This study thus identifies 36% WMS as the optimal replacement level for balancing strength and workability.

The composite mix RH18MD36SF1.5, incorporating 18% RHA, 36% WMS, and 1.5% hooked steel fibers, demonstrated the highest performance, showing a 47.6% increase in compressive strength, a 45.8% rise in split tensile strength, and a 49.6% improvement in flexural strength. Beyond its mechanical advantages, this mix also exhibited significant environmental benefits. An environmental impact assessment (EIA) was performed using the ReCiPe life-cycle impact assessment methodology, renowned for its dual-layered approach with midpoint and endpoint analyses. The midpoint analysis examines specific environmental impacts like particulate emissions, greenhouse gas emissions, and fossil fuel depletion, while the endpoint analysis evaluates broader impacts on human health, ecosystem quality, and resource availability. This dual focus of the ReCiPe method provides a more holistic view, allowing researchers to identify both specific environmental challenges and their cumulative effects on sustainability.

For the RH18MD36SF1.5 mix, the ReCiPe analysis showed a 36.5% reduction in particulate emissions, a 36.6% decrease in greenhouse gas emissions from fossil fuel use, and a 41.7% lower fossil fuel consumption compared to conventional concrete. These environmental improvements stem from reduced cement content, minimized use of natural aggregates, and the sustainable incorporation of agricultural and industrial waste materials. This approach aligns with the UN Sustainable Development Goals (SDGs), particularly SDG 9, which encourages resilient infrastructure and innovation, and SDG 12, which advocates for responsible consumption and waste reduction. This study highlights how utilizing waste as a valuable resource in construction can contribute to sustainable development, mitigate environmental impacts, and foster innovative solutions for eco-friendly infrastructure. This promotes both a circular economy and environmental responsibility within the construction sector.

Reinforcement placement adjustments in RHA and WMS-enhanced concrete focus on ensuring optimal bond and corrosion resistance. Due to the denser matrix from RHA's pozzolanic reaction and WMS's filler effect, precise rebar spacing is crucial to allow effective compaction around reinforcement, minimizing voids and enhancing load transfer. Careful placement and consolidation techniques, such as mechanical vibration

near reinforcement, are essential to maintain strong bonding at the interface, while reduced permeability around rebar from these materials offers added protection against corrosion. These adjustments ensure structural integrity and longevity in reinforced concrete applications.

Data availability statement

All data is included in the manuscript. The data that support the findings of this study are available upon reasonable request from the authors.

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